

TECHNICAL REPORT

Bioremediation and Biodegradation

Evaluation of manure production and characteristics in a beef cattle bedded pack barn: Effects of season and growth stage

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Abstract

Accurate estimates of daily manure production per animal are essential for designing livestock facilities and developing environmental policies. This study assessed manure production in bedded pack barns, a common type of beef cattle housing in Korea, where excreted manure is mixed with bedding and composted before removal. Because composting alters manure mass through moisture evaporation and organic matter decomposition, manure production was evaluated by season (summer, autumn, winter) and cattle growth stage (growing calf, steer, cow). A reliable baseline was established by directly measuring raw manure excretion in a pen for 12 days under controlled conditions (average temperature of -3.9°C and a wind speed of 0.1 m/s), under which moisture evaporation and organic matter decomposition were negligible. Average manure production varied by growth stage: 11.8 ± 2.9 kg/head/day for growing calf, 11.2 ± 2.7 kg/head/day for steer, and 15.8 ± 4.7 kg/head/day for cow. Manure production showed a significant correlation with feed intake ($R^2 = 0.69$), which partially explained the variation across growth stages; however, composting processes influenced by seasonal factors also played important roles. Compared to the baseline, moisture mass decreased by 50%–55% in summer, 12%–22% in autumn, and 6%–21% in winter. Volatile solids decomposition ranged seasonally from 37% to 48% in summer to 9% to 14% in winter, with autumn values between 22% and 28%. Consequently, total manure production declined by approximately 51% in summer and 18% in winter relative to the baseline. These findings underscore the importance of considering seasonal effects, growth stages, and composting practices when estimating manure production in bedded pack barns.

Plain Language Summary

Estimating daily manure production is essential for designing facilities and plays a crucial role in farm management. In Korea, beef cattle are commonly raised in

Abbreviations: ANCOVA, analysis of covariance; BW, body weight; d.b., dry basis; DMI, dry matter intake; MC, moisture content; OM, organic matter; VS, volatile solids; w.b., wet basis; WHC, water-holding capacity.

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bedded pack barns, where manure mixes with bedding and composts over time. This study quantified manure production across seasons (summer, autumn, and winter) and growth stages (calves, steers, and cows), while assessing composting effects in bedded pack barns by accounting for moisture evaporation and organic matter decomposition. Manure production varied by growth stage and season, with cows generating the most (15.8 kg/head/day), followed by growing calves (11.8 kg/head/day) and steers (11.2 kg/head/day). Summer showed the greatest reductions due to moisture evaporation (50%–55%) and organic matter loss (37%–48%), resulting in a 51% reduction compared to baseline manure production. Seasonal and growth-stage effects on manure mass, along with composting losses, suggest potential nutrient loss and pollutant emissions, such as ammonia.

1 | INTRODUCTION

Manure production refers to the typical amount of manure determined by an arithmetic mean of values reported in reference data for each animal species (ASAE, 2005). In these references, manure production is generally measured as the mass of urine and feces based on animal species, growth stages, age, and productivity in the metabolic chamber with diets designed to meet energy and nutrient requirements. While these data serve as a standard for designing housing, manure collection and storage facilities, manure treatment systems, as well as for developing manure management policies and regulations, their actual values vary significantly due to factors such as animal genetics, diet, feeding programs, management practices, and environmental conditions at the site (ASAE, 2005). Therefore, the American Society of Agricultural Engineers (ASAE, 2005) recommends considering such information whenever site-specific data are available rather than relying solely on the average values presented in their standards. However, it is not feasible to measure and use site-specific data due to the intensity of labor, time, and cost. There is a recognized need to investigate manure production under representative management practices and housing conditions, rather than relying on rule-of-thumb estimates.

To better account for housing- and management-driven changes, the ASAE (2005) update introduced two categories of manure production: the “as excreted type,” representing immediate excretion, and the “as removed type,” accounting for changes in manure production due to housing type and management practices (ASAE, 2005). While the “as removed type” of manure production for representative management types has been investigated, for example, in the case of beef cattle, manure production at a feedlot, there is still a need for further research on various housing and manure management systems.

A bedded pack barn, or compost-bedded pack barn, is an alternative and versatile low-cost loose housing system that

utilizes crop residuals, sawdust, and wood shavings as bedding (Honeyman, 2005; Janni et al., 2007). This system has been widely adopted for housing cattle, particularly dairy cows, due to its advantages in improving animal welfare, supporting livestock productivity, and facilitating manure management (R. A. Black, 2013; R. Black, Bewley, et al., 2013; R. Black, Taraba, et al., 2013; Eckelkamp, 2014; Eckelkamp et al., 2016; Leso et al., 2020; Wagner, 2002). However, bedded pack barns for beef cattle are mostly used in regions with relatively high annual precipitation, particularly in the quad-state region of Iowa, Minnesota, South Dakota, and Nebraska, where they are typically operated as deep-bedded confinement barns (Spiehs et al., 2011, 2013; Spiehs, Brown-Brandl, Berry, et al., 2014). These systems have been adopted because they reduce the potential for feedlot runoff and offer advantages in labor efficiency, manure management, and animal performance compared with open-lot feedlots (Doran et al., 2010; Honeyman et al., 2008). A key feature of the bedded pack barn for both dairy and beef cattle is that the excreted manure is mixed with the bedding material and composted until the mixture is removed (Leso et al., 2020). During the composting process, moisture content (MC) evaporation and organic matter (OM) decomposition occur, which can reduce the overall amount of manure (R. A. Black, 2013; Eghball et al., 1997; Michel et al., 2004). The extent of composting varies based on internal and external factors, such as manure management practices and seasonal conditions (Bjerg & Klaas, 2014; R. Black, Bewley, et al., 2013). While several studies have focused on effective bedding management strategies to maximize composting and environmental impacts of bedded pack barns (Bewley & Black, 2012; R. A. Black, 2013; R. Black, Bewley, et al., 2013; R. Black, Taraba, et al., 2013; Doran et al., 2010; Eckelkamp et al., 2016; Leso et al., 2020; Spiehs et al., 2011, 2013; Spiehs, Brown-Brandl, Parker, et al., 2014), no studies have investigated the change in manure production mass due to composting within the bedded pack barn system.

This study evaluated as removed manure production and manure characteristics across different seasons (summer, autumn, and winter) and growth stages (growing calf, steer, and cow) in a beef bedded pack barn. We hypothesized that manure production and characteristics in a bedded pack barn differ by season and growth stage. Specifically, we expected that warmer seasonal conditions would result in lower manure production from the bedded pack barn due to enhanced composting driven by higher temperatures and adequate MC conditions of bedded packs, and that manure production would also vary among growth stages as a consequence of growth stage-dependent differences in dry matter intake (DMI) and nutrient utilization. To test these hypotheses, manure production was directly assessed through weight measurements and analyses of MC evaporation and degree of volatile solids (VS) decomposition. To compare variation in manure production by season and growth stage, we measured a baseline of manure production, which represents the amount of manure excreted by cattle at each growth stage in a bedded pack barn under conditions that minimize MC evaporation and OM decomposition. By combining direct manure mass measurements and baseline manure metrics, this study elucidates the influence of seasonal environment and growth stages on manure production in a bedded pack barn. Importantly, the findings provide quantitative evidence of season- and growth stage-specific differences in manure production, thereby addressing a key knowledge gap in manure management for bedded pack barn systems.

2 | MATERIALS AND METHODS

2.1 | Bedded pack barn, animal, and diet

This study was conducted at the Chungnam National University farm (Cheongyang-gun, South Korea). The concrete floor with bedding and a mound of manure (urine and feces), called a bedded pack, represents a typical Korean bedded pack barn (Figure 1) and operates as a loose housing system equipped with mechanical ventilation fans (diameter: 1 m, maximum airflow: 350 m³/min). Detailed characteristics of a typical Korean bedded pack barn are summarized in the [Supporting Information](#). These fans were installed at a 45° angle, 3 m above the ground, with one fan per four pens (Figure 1; Figure S1). The ventilation fans operated at 30% of their maximum airflow when the summer temperature exceeds 28°C, 50% at 30°C, and 100% (maximum airflow) at 32°C or higher. In winter, the sidewall curtains were lowered to the floor to block outside wind.

Twenty-one beef cattle (breed: Hanwoo beef cattle) were divided into three groups based on their growth stage: growing calf, steer, and cow. The cattle were housed in nine pens (Figure 1), each with an area of 25 m², over three seasons:

Core Ideas

- Average manure production was 11.8 ± 2.9 kg/head/day (calf), 11.2 ± 2.7 kg/head/day (steer), and 15.8 ± 4.7 kg/head/day (cow).
- Moisture loss was 50%–55% in summer, 12%–22% in autumn, and 6%–21% in winter compared to the baseline.
- Volatile solids loss was 37%–48% in summer, 22%–28% in autumn, and 9%–14% in winter compared to the baseline.
- Total manure production was lower by ~51% in summer and ~18% in winter compared to the baseline.

summer (85 days), autumn (77 days), and winter (67 days). Stocking densities (Table S1) were determined according to the Handbook of Agricultural Technology (RDA, 2015) and suggested optimal animal densities (Leso et al., 2020) in compost bedded pack barns. Accordingly, growing calves were housed at a density of 8.3 m²/head (three heads per pen), while steers and cows were housed at 12.5 m²/head (two heads per pen). Cattle with similar body weight (BW) were grouped and placed in the same pen within each growth stage.

All animals were fed a total mixed ration (TMR) diet consisting of timothy hay and a commercial concentrate mix (CM). The TMR ratio was 800 g/kg timothy hay and 200 g/kg CM. The chemical compositions of the timothy hay and CM are presented in Table S2. The diet was provided ad libitum and fed twice daily at 8:00 a.m. and 5:00 p.m. Animals had free access to drinking water throughout the experiment.

Each pen was equipped with an electronic feed bin that automatically measured individual DMI by using radio-frequency identification tags attached to the animals (Dawoon Co.). The BW of cattle was measured at the beginning and end of the experiment using a scale (Gwangjin Co. Ltd.). The results of the performance by growth stage during the experimental period are shown in Table 1.

2.2 | Bedding material

Based on a survey of 120 Korean cattle farms nationwide (HWB, 2019), sawdust was identified as the most commonly used bedding material. Given its widespread use in the industry, sawdust was selected as the bedding material for this study. The characteristics of the sawdust used in this study are shown in Table S3. It was added to the pens to a thickness of 5 cm (290–292 kg/pen, depending on bulk density), following the standard design for livestock manure resource utilization in Korea (RDA, 2008). The manure production (bedded pack;



FIGURE 1 The typical bedded pack barn in Korea. The red dotted box indicates the woody partition (5 m × 0.4 m × 0.05 m) installed to prevent manure overflow and ensure accurate manure mass determination in each pen.

bedding-manure mixture) was removed and measured when the MC reached saturation and manure began to stain the cattle's udders and thighs, consistent with criteria described in the previous research (R. A. Black, 2013; Eckelkamp, 2014; Jo et al., 2017). Under these criteria, the bedded pack was removed after 85 days in summer, 77 days in autumn, and 67 days in winter in this study. After bedded pack removal, the pens were replenished with fresh bedding for the subsequent trial.

2.3 | Seasonal data collection

The air temperature and relative humidity inside the barn were measured hourly using an Arduino Uno with a DHT22 sensor (AM2302, Arduino). Five DHT22 sensors were installed in each hallway in the barn. Seasonal averages of air temperature and relative humidity inside the barn showed the highest values in summer (average: 23°C and 90%) and the lowest in winter (average: 0.3°C and 81%), with detailed numerical ranges presented in Table S4.

2.4 | Manure sampling

To investigate the physicochemical properties (MC and VS) of the manure production in the bedded pack barn, samples were collected every 7 days for the first 30 days over each season. Afterward, samples were collected at 2-week intervals. The sampling method involved evenly mixing the bedded pack using manual labor and an electric tiller (Greenworks).

Samples were then collected from multiple points within the barn (Figure S2).

2.5 | Measurement of manure production

In this study, manure production was defined as the mass of the manure “as removed type” throughout the manuscript. In bedded pack barns, excreted feces and urine are immediately absorbed and mixed with the bedding material, forming a bedded pack (bedding-manure mixture). Accordingly, as-removed manure production was quantified by measuring the mass of the bedded pack at the end of each experimental period and subtracting the mass of the initially applied bedding material. Throughout each experimental period, no runoff, leachate loss, or overflow of manure between adjacent pens was observed, allowing accurate measurement of the bedded pack mass for each pen.

To measure bedded pack mass, the bedded pack removed from each pen was loaded onto a trailer equipped with a load cell scale (resolution: 20 kg; accuracy: $\pm 1\%$ – $\pm 2\%$; RW-10Z, CAS Scale Korea). The net mass of the bedded pack was calculated by subtracting the trailer tare weight from the gross weight (Figure S3). Daily manure production per head was calculated by first subtracting the mass of the initial bedding material (290–292 kg) from the net bedded pack mass to obtain the net manure production mass. This value was then divided by the number of experimental days (85 days in summer, 77 days in autumn, and 67 days in winter) and by the number of cattle housed in each pen (two or three heads) to estimate daily manure production per head.

TABLE 1 Performance of experimental cattle.

Items	Summer			Autumn			Winter		
	Growing calf	Steer	Cow	Growing calf	Steer	Cow	Growing calf	Steer	Cow
Initial BW (kg)	205.3 ± 29.1	607.8 ± 23.9	532.7 ± 43.4	180.8 ± 14.1	654.5 ± 34.8	537.8 ± 77.5	184.9 ± 18.2	660.1 ± 37.4	575.8 ± 65.1
Final BW (kg)	257.7 ± 37.3	656.1 ± 35.8	575.8 ± 55.1	259.7 ± 18.9	710.8 ± 22.4	561.3 ± 49.9	257.1 ± 24.0	687.0 ± 48.7	596.2 ± 67.8
DMI (kg/head/day)	7.1 ± 1.6	7.2 ± 1.5	10.4 ± 1.1	8.1 ± 2.4	7.3 ± 0.6	12.6 ± 0.8	9.5 ± 3.6	7.8 ± 0.5	12.0 ± 1.9
ADG (kg/head/day)	0.8 ± 0.4	0.7 ± 0.3	0.6 ± 0.3	0.8 ± 0.1	0.6 ± 0.2	0.2 ± 1.3	0.9 ± 0.1	0.4 ± 0.89	0.1 ± 0.5

Note: *N* (number of samples) of growing calf = 9; *N* of steer and cow = 6; the reported value in the table = mean ± standard deviation.

Abbreviations: ADG, average daily gain; BW, body weight; DMI, dry matter intake.

2.6 | Baseline of manure production

A baseline for Hanwoo manure production was required to evaluate the seasonal and growth stage variation of MC evaporation, VS decomposition, and manure production in the bedded pack barns. While the baseline of manure production could be determined using the metabolic chamber (12.85 kg/head/day) as a standard reference provided by the Ministry of Environment Provisions (MEP, 1999, 2022) in South Korea, we recognize that measurements obtained from the metabolic chamber have limitations when compared to those observed in the bedded pack barns. These limitations arise from differences in animal behavior and feeding conditions (such as feed ingredients, feed type, DMI, and stocking density) between metabolic chambers and bedded pack barn environments. Therefore, we established our own baseline of manure production by directly measuring in the bedded pack barn (MPb) when MC evaporation and OM decomposition are minimized over a short period of winter. The MPb was measured for 12 days of winter (average temperature of -3.9°C) in a pen, where the inflow of external wind was minimized by winch curtains for each growth stage (growing calf, steer, and cow). The MPb after 12 days was estimated by directly measuring the weight of manure production using manual labor and scales. During the period in which MPb was measured, MC evaporation and VS decomposition within the bedded pack were closely monitored (Figure S4), and no statistically significant reductions were observed (Table S5). Detailed descriptions of the monitoring procedures are provided in the Supporting Information.

2.7 | Sample analysis: Moisture content, volatile solids, and water-holding capacity

Manure samples collected for each season and growth stage were stored at 4°C immediately after collection and analyzed for manure characteristics within 48 h. The MC was measured based on weight loss after drying the sample at 105°C for more than 1 day, and VS were measured using the weight difference before and after incineration at 550°C for 8 h (USCC, 2002). The moisture mass or VS mass of the manure production was calculated by multiplying the end-point MC or VS (%) of the bedded pack by the final mass of the bedded pack. Then, the mass of MC or VS originating from the initial bedding was removed. Water-holding capacity (WHC) refers to the maximum MC that a material can absorb, and by measuring this, it is possible to estimate the maximum MC that does not generate leachate. WHC was analyzed by modifying the Hilgard method (Ahn et al., 2008; Mabuhay et al., 2003).

2.8 | Statistical analysis

All statistical analyses were conducted using R Studio (programming language R). Because manure production could not be measured at the individual animal level within a pen, the pen was defined as the experimental unit for all manure production-related analyses. For example, DMI and BW were measured at the individual animal level; however, for analyses examining their relationships with manure production, individual measurements were averaged within each pen to ensure consistency with the pen-level manure production data. The manure production was measured in three pens (triplicate) for each combination of season (summer, autumn, winter, and baseline) and growth stage (growing calf, steer, and cow), resulting in a total of 36 pen-level observations ($n = 36$).

One-way analysis of variance (ANOVA) was first performed to determine the statistical significance of seasonal and growth stage manure production, and the Tukey test was conducted as a post-hoc test ($p \leq 0.05$). Before using the ANOVA analysis, the parametric assumptions were checked by the Brown–Forsythe test (homogeneity of variance) and the Shapiro–Wilk test (normal distribution of data).

In addition, a linear regression analysis was used to examine associations between manure production and potential explanatory variables, including DMI, BW, air temperature, and relative humidity. Environmental variables (air temperature and relative humidity) were summarized as average values for each season up to the day on which manure production was measured and were used to evaluate correlations with manure production. When manure production of different growth stages was measured on the same day, environmental data were averaged only up to the hourly time of measurement and included accordingly. Model significance was evaluated at $p \leq 0.05$, and the coefficient of determination (R^2) was reported.

To test for differences in manure production among seasons and growth stages while accounting for the strong association between manure production and DMI, we then fit an analysis of covariance (ANCOVA) model with seasons and growth stages as fixed factors and DMI as a continuous covariate (Equation 1). When the ANCOVA indicated significant main effects or interactions, post-hoc comparisons were performed using Tukey's HSD test ($p \leq 0.05$).

$$Y_i = \mu + \alpha_{S(i)} + \gamma_{G(i)} + \beta \text{FI}(i) + \epsilon, \quad (1)$$

where Y is the manure production for observation i , μ is the intercept, α is the fixed effect of season, $S(i)$ is the season indicator (dummy) variables for observation i , γ is the fixed effects for growth stage, $G(i)$ is the growth-stage indicator (dummy) variables for observation i , β is the covariate slope, $\text{FI}(i)$ is

the feed intake, expressed as DMI, for observation i , and ϵ is the error term.

3 | RESULTS

3.1 | Defining a baseline of manure production

Baseline for manure production in a bedded pack barn (MPb) was quantified under controlled conditions to minimize the influence of MC evaporation and VS decomposition (Table S5). During the period of MPb measurement, the average DMI of the cattle was 11.0 ± 2.7 kg for growing calves, 8.8 ± 0.6 kg for steers, and 13.6 ± 2.0 kg for cows. DMI during the MPb period did not differ significantly from that measured over the full winter period ($p > 0.05$); however, it differed significantly from that measured during the summer and autumn periods ($p < 0.05$), with the exception of the autumn cow group.

Baseline manure production values were 15.8 ± 3.6 kg for growing calves, 15.3 ± 2.0 kg for steers, and 19.7 ± 3.1 kg for cows (Table 2). MC (wet basis; w.b.) exhibited a decreasing trend from growing calf ($85.3 \pm 3.4\%$) to steer ($82.9 \pm 1.8\%$) and cow ($81.2 \pm 1.7\%$). In contrast, VS (dry basis; d.b.) remained relatively stable across growth stages, ranging from $82.0 \pm 0.1\%$ to $82.9 \pm 1.8\%$.

3.2 | Manure production by season and growth stage

Across all growth stages, manure production was highest for baseline (MPb), followed by winter, autumn, and summer (Table 3). Compared to MPb, manure mass decreased by 46%–51% in summer, 11%–22% in autumn, and 6%–18% in winter. Manure production during the summer was significantly lower than in other seasons across all growth stages ($p = 0.0002$). However, there were no significant differences in manure production between MPb and the autumn–winter seasons.

Manure production varied among growth stages ($p = 0.03$), with cows producing the most, followed by growing calves and then steers. A significant difference was observed between cows and steers.

Although BW was highest in steers (607–660 kg), followed by cows (533–576 kg) and growing calves (181–205 kg), manure production was not significantly correlated with BW ($p > 0.05$). In contrast, manure production was strongly associated with DMI, with a coefficient of determination (R^2) of 0.69 ($p < 0.0001$; Figure S5). No significant relationship was found between manure production and the other environmental factors (air temperature and relative humidity). These

TABLE 2 Baseline manure production and corresponding moisture and volatile solids contents evaluated in the bedded pack barn (MPb).

Items	Growing calf	Steer	Cow
DMI (kg/head/day)	11.0 ± 2.7	8.8 ± 0.6	13.6 ± 2.0
BW (kg)	184.9 ± 18.2	660.1 ± 37.4	575.8 ± 65.1
Manure production (kg/head/day)	15.76 ± 3.58	15.26 ± 2.02	19.74 ± 3.11
Water in manure (kg)	13.53 ± 3.43	12.67 ± 1.89	16.07 ± 2.80
Total solids in manure (kg)	2.23 ± 0.15	2.59 ± 0.12	3.67 ± 0.33
Volatile solids in manure (kg)	1.83 ± 0.12	2.13 ± 0.10	3.04 ± 0.27

Abbreviations: BW, body weight; DMI, dry matter intake.

TABLE 3 Manure production by season and growth stage in a bedded pack barn.

Growth stage	Items (unit: kg/head/day)	Baseline (MPb)	Summer	Autumn	Winter
Growing calf	Manure production	15.8 ± 4.4b	8.5 ± 2.2a	12.6 ± 0.7b	13.5 ± 4.1b
	Water in manure	13.5 ± 4.2b	6.7 ± 1.8a	10.7 ± 0.5b	11.4 ± 3.5b
	Volatile solids in manure	1.8 ± 0.2b	1.1 ± 0.3a	1.4 ± 0.1a	1.6 ± 0.5ab
Steer	Manure production	15.3 ± 2.5b	8.1 ± 1.9a	12.0 ± 2.8b	12.5 ± 0.7b
	Water in manure	12.7 ± 2.3b	6.3 ± 1.6a	9.8 ± 2.6ab	9.9 ± 0.9ab
	Volatile solids in manure	2.1 ± 0.1b	1.2 ± 0.3a	1.5 ± 0.2ac	1.9 ± 0.2bc
Cow	Manure production	19.7 ± 3.8b	9.6 ± 1.0a	17.5 ± 1.1b	18.4 ± 6.4b
	Water in manure	16.1 ± 3.4b	6.8 ± 1.2a	14.1 ± 1.0b	15.1 ± 5.2b
	Volatile solids in manure	3.0 ± 0.3b	1.9 ± 0.4a	2.3 ± 0.1ab	2.5 ± 1.0ab

Note: Values within a row with different lowercase letters differ significantly at $p < 0.05$.

findings indicate that DMI is a primary driver of variation in manure production.

ANCOVA was applied to test whether manure production differed by season and growth stage after adjusting for DMI; then it was identified as a significant covariate ($p < 0.0001$), indicating that manure production increased significantly with higher DMI, consistent with the high R^2 observed in the regression analysis. Even after controlling for DMI, significant differences in manure production were observed across seasons ($p = 0.001$; between summer and other seasons) and growth stages ($p = 0.018$; between cow and steer).

In conclusion, based on an annual cycle consisting of 3 months of summer, 6 months of autumn and spring (autumn values were used to represent spring), and 3 months of winter (e.g., [three × summer mean + six × autumn mean + three × winter mean]/12), the average daily manure production per head of Hanwoo was estimated to be 11.8 ± 2.9 kg for growing cattle, 11.2 ± 2.7 kg for steer, and 15.8 ± 4.7 kg for cow.

3.3 | The change in moisture content

The MC of the bedded pack in bedded pack barns showed a rapid increase during the first 15 days (Figure 2). After this initial period, the rate of increase slowed down and

eventually stabilized, regardless of the growth stage and season.

While the trend in MC change of the bedded pack was similar regardless of the growth stage and season, the degree and timing of MC stabilization in the bedded pack varied by season. For steer and cow (Figure 2B,C), MC stabilized at 63.8 ± 9.9% around days 25–30 in the summer. In autumn, the MC stabilized at 69.1 ± 4.3% around day 40, whereas in winter, it stabilized at approximately 69.7 ± 4.0% around days 30–35. In contrast, the bedded pack stabilized MC at 72.1 ± 1.6% after approximately 23 days in the growing calf, regardless of the season (Figure 2A).

Relative to MPb, moisture mass was significantly lower during the summer (50%–55%), autumn (12%–22%), and winter (6%–21%) periods (Table 3; Figure S6).

3.4 | The change in volatile solids

Figure 3 illustrates seasonal changes in VS content of the bedded pack across growth stages. A consistent decrease in VS content was observed, regardless of the growth stage and season, but the magnitude and timing of decline differed among seasons and growth stages.

Specifically, for the growing calf, while reduction rates were similar during autumn and winter, a significant diver-

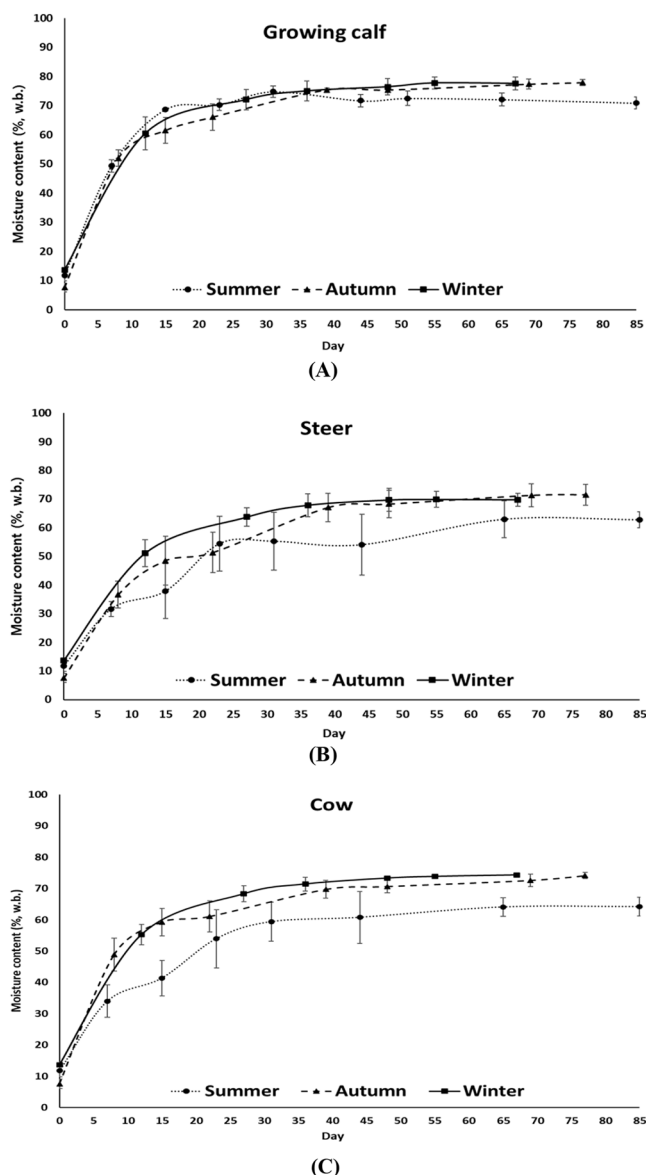


FIGURE 2 Changes in moisture content in the bedded pack: (A) growing calf, (B) steer, and (C) cow. w.b., wet basis.

gence occurred in summer after 15 days, with summer showing approximately 9 percentage points lower VS content. This trend persisted throughout the experiment, resulting in a final VS content of 77% (d.b.) in summer compared to 85% (d.b.) in autumn and winter.

For steers, seasonal differences in VS content were not apparent until day 55. After 65 days, the VS content in summer declined relative to autumn and winter, resulting in final values of about 80% (d.b.) in summer and 85%–86% (d.b.) in autumn and winter.

Cow showed the most pronounced seasonal differences in VS content. Within 10–15 days, VS content was highest in winter, followed by summer and autumn. After 65 days, VS content in summer decreased below that in autumn. At the

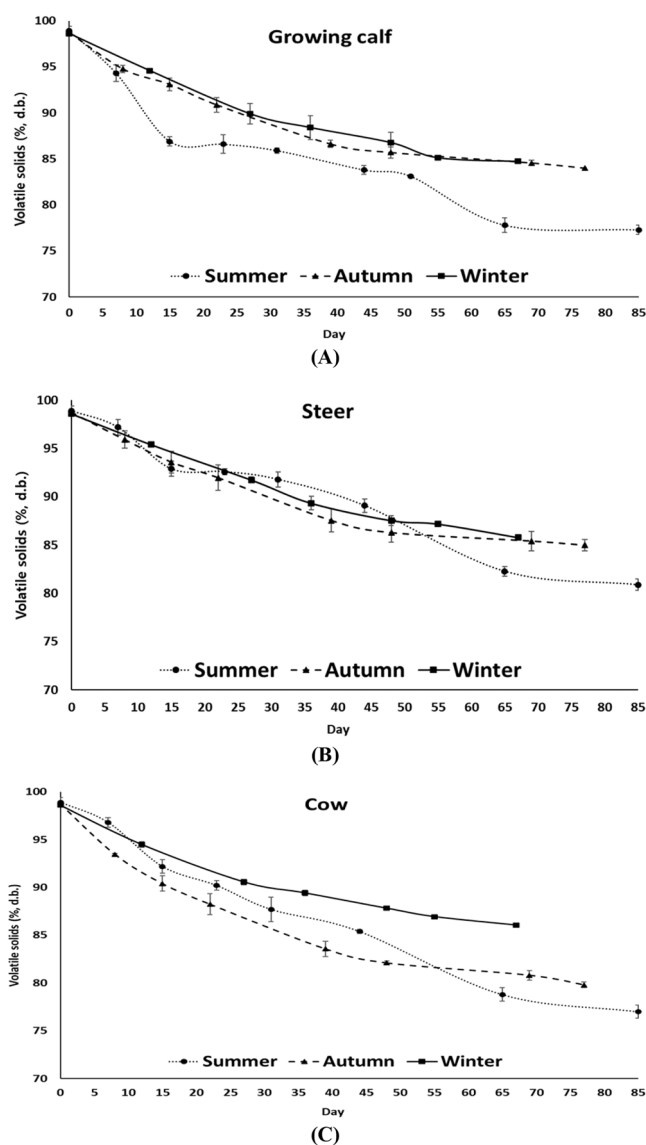


FIGURE 3 Change in volatile solids in the bedded pack: (A) growing calf, (B) steer, and (C) cow. d.b., dry basis.

end of the experimental period, VS content was 87% (d.b.) in winter, 80% (d.b.) in autumn, and 78% (d.b.) in summer.

Compared to the MPb baseline, the VS mass in bedded pack barns was significantly reduced by approximately 37%–48% in summer, 22%–28% in autumn, and 9%–14% in winter (Table 3; Figure S7).

4 | DISCUSSION

4.1 | Considerations on appropriate timing for bedding replacement or removal

As removed manure production and characteristics in bedded pack barns are influenced by multiple interacting factors,

including the initial quantity of bedding material, the continuous accumulation of freshly excreted manure, bedding management practices, and mass losses driven by MC evaporation and OM decomposition (R. A. Black, 2013; Eghball et al., 1997; Michel et al., 2004). From the perspectives of animal welfare, barn environmental management, and farm economics, determining appropriate timing for bedding removal, replacement, or supplementation is critical (Fregonesi et al., 2007; Jo et al., 2017; Tucker & Weary, 2004). Prolonged use of bedding materials can promote vermin infestation and disease, reduce animal productivity, and disrupt aerobic composting through excessive MC accumulation, leading to odor emissions and low-quality compost (Ahn et al., 2008; Das & Keener, 1997; Jo et al., 2017). Accordingly, an appropriate bedding management cycle is essential to balance animal hygiene, manure management efficiency, and operational costs.

The optimal interval for bedding removal or additional bedding input varies with animal density, climatic conditions, ventilation rate, and hygiene requirements (Barberg et al., 2007; Janni et al., 2007). Empirically, bedding removal or replacement is commonly performed when pack MC approaches saturation and visible manure contamination of cattle udders and thighs occurs (R. A. Black, 2013; Eckelkamp, 2014; Jo et al., 2017). A key concern, however, is whether such empirically derived criteria consistently promote appropriate composting conditions and remain applicable across different seasons and growth stages.

In the present study, following bedding application, the MC of the bedded pack increased rapidly during the first 30 days, reaching approximately 60%–75% of the sawdust bedding. After this initial period, MC increased more gradually until the end of the experimental period (Figure 2), indicating that the bedding had reached near-saturation conditions with limited capacity for additional MC absorption. Subsequently, manure removal for quantifying as-removed manure production was conducted when visible manure contamination of the udders and thighs was observed in most cattle within a pen, which occurred approximately 65–85 days after bedding application. These findings suggest that visible manure contamination generally occurred after the bedded pack reached near-saturation MC conditions and persisted for several weeks to approximately 2 months, depending on season and growth stage. Longer bedding use was observed during summer (approximately 80 days), whereas shorter replacement intervals were required during winter (approximately 60 days). Growing calf pens exhibited faster MC saturation relative to other growth stages. Collectively, bedding materials remained usable for approximately 60–80 days before substantial hygiene concerns were observed.

However, from a composting perspective, optimal MC for efficient aerobic composting is typically 50%–60% (Kim

et al., 2015; Liang et al., 2003). MC exceeding this range can fill pore spaces, promote anaerobic conditions, and impair OM decomposition while increasing odor emissions (Ahn et al., 2008). Therefore, although bedding materials may remain in use for extended periods before animal hygiene is compromised, earlier intervention (between 10 and 30 days) through bedding replacement, supplementation, or mechanical mixing may be required to maintain favorable composting conditions.

Overall, these findings suggest that to balance effective manure composting, animal hygiene, and practical bedding materials management may benefit from active management practices, such as replacement, supplementation, or mechanical agitation for bedding, within approximately 10–30 days. In systems typical of Korean bedded pack barns, where bedding materials are removed entirely rather than partially replenished, a management cycle of approximately 40–50 days may represent a realistic compromise for maintaining controllable MC levels and composting performance, when labor requirements, bedding costs, and manure handling constraints are considered. Jo et al. (2017) reported bedding management intervals of approximately 30–50 days in beef cattle bedded pack barns, consistent with the present findings.

4.2 | Influence of season and growth stage on manure production and characteristics

Results from the present study demonstrate that both season and growth stage significantly influence as-removed manure production and characteristics in bedded pack barns. With respect to seasonal effects, manure production was lowest in summer and highest in winter, while manure removed at the end of the experimental period exhibited higher MC in winter and lower MC in summer. These seasonal patterns are consistent with differences in MC evaporation and VS decomposition under contrasting environmental conditions. Warmer temperatures and strong ventilation during summer are conducive to greater MC evaporation and VS degradation, whereas cooler winter conditions limit these processes.

Compared with the baseline, manure production was reduced by approximately 6%–51% across seasons, with the greatest reduction observed in summer. The evaporation of MC was the dominant contributor to this reduction, accounting for approximately 65% of the total mass loss. The magnitude of manure mass reduction fell within the general range of 35%–50% reported for composting systems (Eghball et al., 1997), although it was lower than 50%–80% reduction investigated in the pilot-scale dairy manure composting with turned windrows (Michel et al., 2004) and the approximately 62% reduction observed in bedded pack barns equipped with underfloor aeration (Bjerg & Klaas, 2014). These comparisons suggest that, relative to other seasons, elevated

summer temperatures combined with adequate ventilation can create favorable conditions for enhanced composting activity, thereby promoting OM decomposition and MC evaporation.

Seasonal trends in manure characteristics were generally consistent across growth stages. The MC of the bedded pack increased rapidly during the initial 30 days and thereafter increased more gradually, without exceeding the WHC of the sawdust bedding. This pattern is characteristic of bedded pack barns and has been reported previously (Bjerg & Klaas, 2014; R. Black, Bewley, et al., 2013; Jo et al., 2017). Two possible mechanisms may contribute to this stabilization of MC. First, the bedding material may become saturated with MC and thus cannot absorb any additional MC (Jo et al., 2017). Second, under conditions of high temperatures and increased ventilation, such as during the summer, the rate of MC evaporation may balance with the MC input from manure excretion, leading to a near-equilibrium state.

The VS content of the bedded pack declined over time, with the rate of decline decreasing after approximately 40–50 days. This reduction reflects not only microbial decomposition of OM in manure and bedding, but also a dilution effect resulting from the accumulation of manure with a lower VS content. In the present study, the sawdust bedding had a VS content of approximately 99% (d.b.), whereas cattle manure exhibited a substantially lower VS content of approximately 82% (d.b.). As manure accumulated within the bedding pack, the overall VS content of the mixture decreased accordingly, a pattern that became more pronounced with continued manure addition.

Growth stage also influenced manure characteristics, particularly MC dynamics within the bedded pack. Pens housing growing calves exhibited the most rapid increase in MC, resulting in higher stabilized MC levels compared with those housing steers and cows. This pattern may be related to the relatively higher activity and water consumption of the growing calves (12 months old) compared to the steer and cow (Yang et al., 2016). In addition, although growing calves had lower BW, they produced substantial quantities of manure (11.8 kg/day), with MC reaching approximately 85% (w.b.), compared with 81%–83% for steers and cows. Consequently, the bedded pack in pens housing growing calves stabilized at a higher MC (approximately 72% w.b.) than that observed for steers and cows (approximately 64% during summer and 69%–70% during autumn and winter).

Interestingly, reductions in VS mass were more pronounced for steers and cows than for growing calves. This difference may be related to the lower stabilized MC in pens housing steers and cows, which may have favored aerobic conditions and enhanced VS decomposition. In contrast, the higher MC observed in growing calf pens may have constrained aerobic composting processes, resulting in comparatively lower VS decomposition.

Overall, these findings indicate that manure production and manure characteristics in bedded pack barns are jointly influenced by seasonal conditions and cattle growth stage. Accordingly, both factors should be explicitly considered when interpreting, comparing, and predicting as-removed manure production in bedded pack barn systems.

4.3 | Limitations of the study

Several limitations of this study should be acknowledged to facilitate appropriate interpretation of the findings and to inform future research directions. First, the influence of the sawdust bedding material on manure production estimates and manure characteristics in bedded pack barns was not fully quantified. In this study, the initial bedding mass was excluded from manure production estimates, as it does not originate from animal excretion, and its decomposition was assumed to be negligible. However, the bedding material may undergo partial decomposition during composting, potentially affecting the estimation of manure mass. In addition, the contributions of manure and bedding to MC and VS in the bedded pack were not independently separated. Although the influence of manure is expected to increase over time as it accumulates, this assumption was not directly quantified and may introduce uncertainty in the interpretation of MC and VS dynamics. Thus, future studies should explicitly quantify the contribution of bedding to improve the accuracy of manure production estimates and characterization in bedded pack barn systems.

A second limitation is that manure production estimates in this study assumed that all feces and urine excreted by the animals were retained within the bedded pack. In practice, a portion of urine may be lost through evaporation before being fully absorbed into the bedding, feces, or the bedded pack. Such unaccounted evaporative losses may influence the reported manure production values. Therefore, future studies should consider quantifying urine evaporation losses to improve the accuracy of manure production estimation.

Addressing these limitations will enhance future experimental designs and improve the robustness and applicability of manure production estimates in bedded pack barn systems.

5 | CONCLUSION

Bedded pack barns offer benefits for animal welfare, productivity, and manure management by mixing excreted manure directly with bedding, where OM decomposition and MC evaporation occur, in a process of composting. This composting process, influenced by barn environment and management, can reduce manure mass. Thus, this study aimed to evaluate manure production by season (summer, autumn,

and winter) and growth stage (growing calf, steer, and cow), investigating the magnitude of MC evaporation and VS decomposition. Compared to the baseline, the results of the study showed that the water mass in manure decreased by 50%–55% in summer, 12%–22% in autumn, and 6%–21% in winter. The decomposition of VS also showed the highest in summer and the lowest in winter, at 37%–48% in summer, 22%–28% in autumn, and 9%–14% in winter. Reflecting these results, the manure production by season was lower in summer and higher in winter (6%–51%). Differences in manure production by growth stage were highly related to DMI ($R^2 = 0.69$). This indicates that it is necessary to estimate the manure production by accounting for growth stage, housing type, season, and manure management to accurately utilize manure sources.

AUTHOR CONTRIBUTIONS

Myeongseong Lee: Data curation; formal analysis; investigation; visualization; writing—original draft; writing—review and editing. **Seunghun Lee:** Data curation; formal analysis; investigation; methodology; writing—review and editing. **Eunjong Kim:** Data curation; formal analysis; investigation; writing—review and editing. **Jisoo Wi:** Data curation; formal analysis; investigation; writing—review and editing. **Heekwon Ahn:** Conceptualization; funding acquisition; investigation; methodology; project administration; resources; supervision; validation; writing—original draft; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data presented in this study are available on request from the corresponding author.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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